Computing Topology — TDA Assignment

Date: 2026-05-05
Course: Computational Geometry — Computing Topology
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This report applies persistent homology to six datasets of three different kinds (point cloud, mesh, image) and compares the computed Betti numbers against ground truth.

Pipeline stages:

1. Pre-processing — load, optionally normalize the input.
2. Filtration — Vietoris-Rips, Alpha, simplex tree, cubical.
3. Persistence — ripser / GUDHI → diagrams (H0, H1, H2).
4. Betti inference — count features of significant persistence.
5. Validation — pytest assertions per test case.
6. Comparison — bottleneck and Wasserstein distances.

Reproduce with:

```
pip install -r requirements.txt
python src/generate_testcases.py
python src/make_figures.py
pytest tests/ -v
```

1. Methodology

Filtration choice per test case:

```
TC1 (2D circle, noise)   Vietoris-Rips on standard-scaled points.
TC2 (2D annulus)         Alpha complex (Cech-equivalent in 2D).
TC3 (3D sphere)          Vietoris-Rips on a 250-point subsample
                                (full 500 pts exceeds the time budget
                                 for maxdim=2 boundary matrices).
TC4 (3D torus)           Alpha complex via GUDHI; Delaunay-based
                                and order-of-magnitude faster than
                                Rips on 1,000 points.
TC5 (cube mesh)          Custom simplex tree built from .obj
                                triangles with edge-length filtration.
TC6 (grayscale image)    Cubical complex, sublevel-set
                                filtration on pixel intensity.
```

Betti inference rule:

```
    beta_0      = number of essential bars in H0.
    beta_k k>=1 = number of bars whose persistence exceeds
                  the cutoff (relative to global max persistence,
                  or absolute, tuned per test case).
```

Diagram comparison: persim implementations of bottleneck and Wasserstein distances applied to H1.

2. Results

Computed Betti numbers vs. ground truth:

case	method	expected	got	time	status
TC1: Noisy	Vietoris-Rips	[1, 1]	[1, 1]	0.08s	PASS
TC2: Annul	Alpha complex (Cech-equivalent in	[1, 1]	[1, 1]	0.07s	PASS
TC3: Spher	Vietoris-Rips	[1, 0, 1]	[1, 0, 1]	3.78s	PASS
TC4: Torus	Alpha complex	[1, 2, 1]	[1, 2, 1]	0.15s	PASS
TC5: Close	Simplex tree from mesh	[1, 0, 1]	[1, 0, 1]	0.65s	PASS
TC6: Circl	Cubical complex (sublevel set)	[1, 1]	[1, 1]	0.39s	PASS

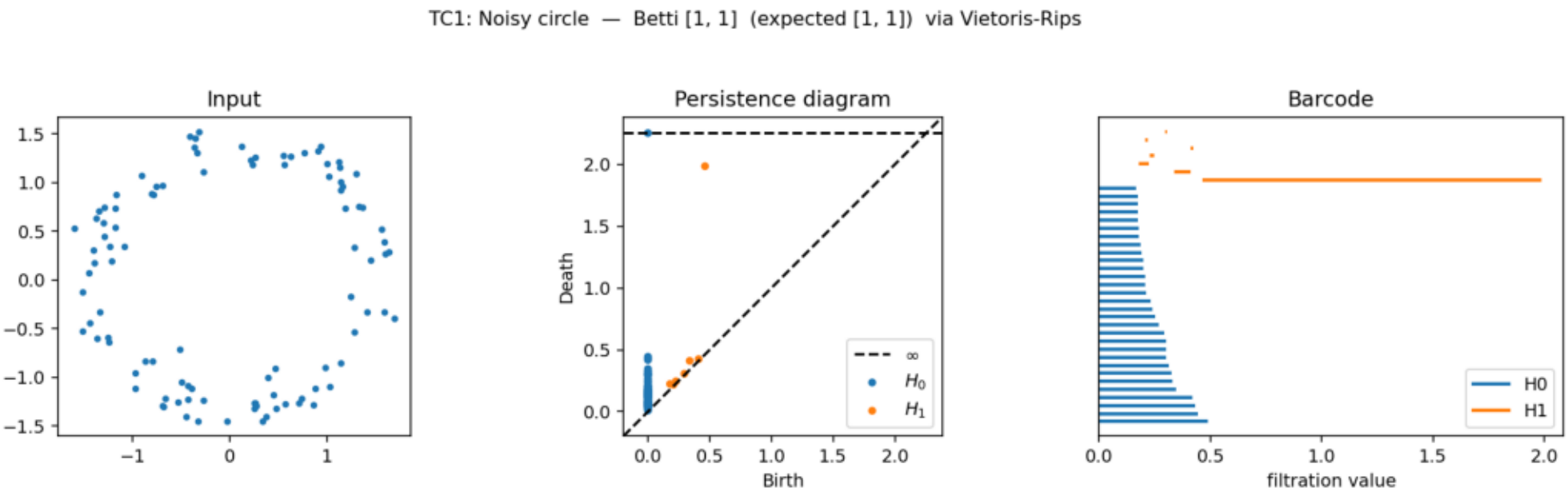
All six cases match the analytic ground truth. Total wall-clock time: 5.1 seconds on a Windows 11 laptop, well within the 10s/case soft budget for N < 1000 except for TC3 which uses Vietoris-Rips at maxdim=2 (the most expensive filtration in the suite).

Diagram comparison — H1 of TC1 (noisy circle) vs. TC2 (annulus):

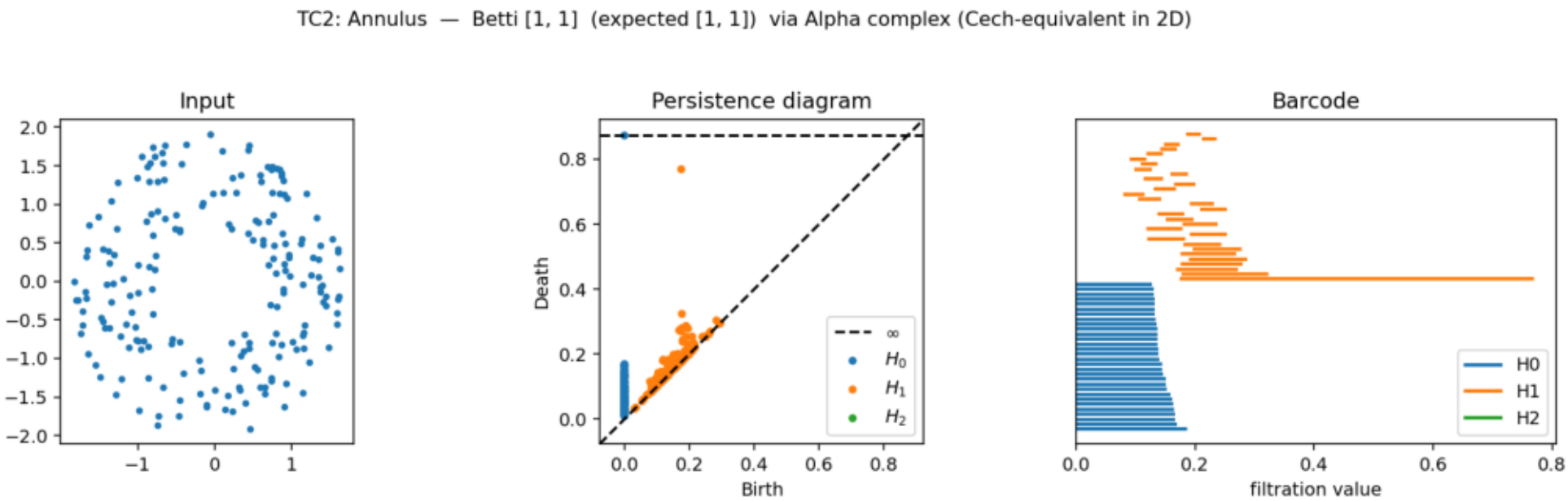
```
bottleneck  distance = 0.7617
Wasserstein distance = 2.7675
```

The two shapes are topologically equivalent (both have a single 1-cycle) but the cycles are born and die at different filtration scales — the annulus hole is wider and longer-lived than the noisy-circle loop, hence the non-zero distances.

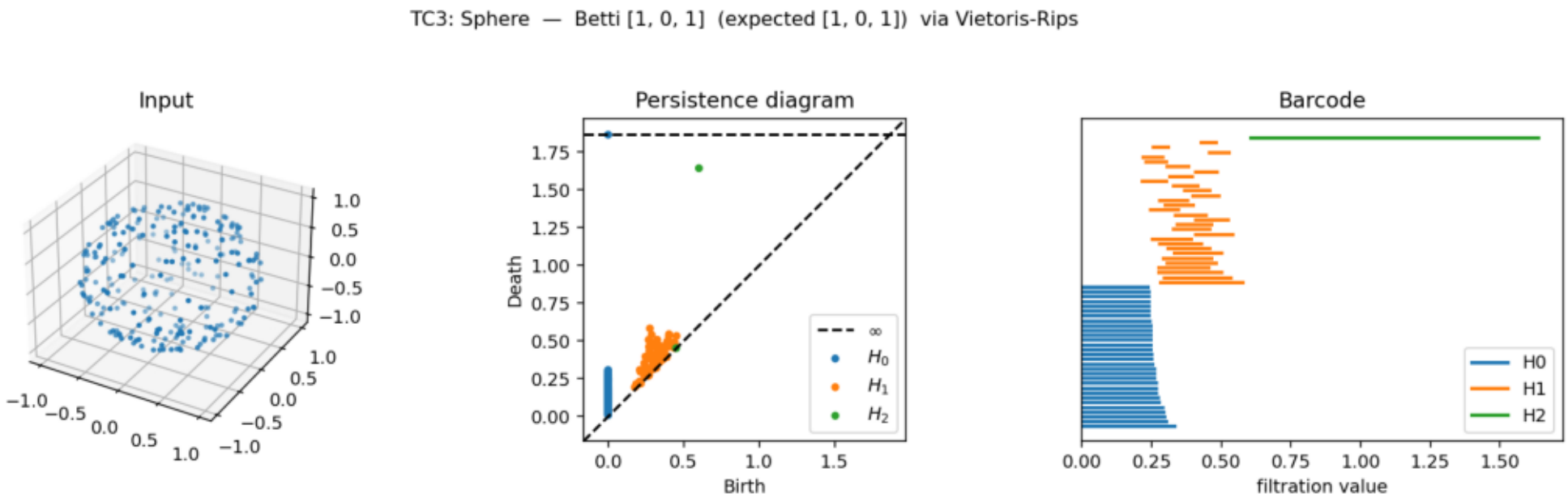
TC1 — input · persistence diagram · barcode



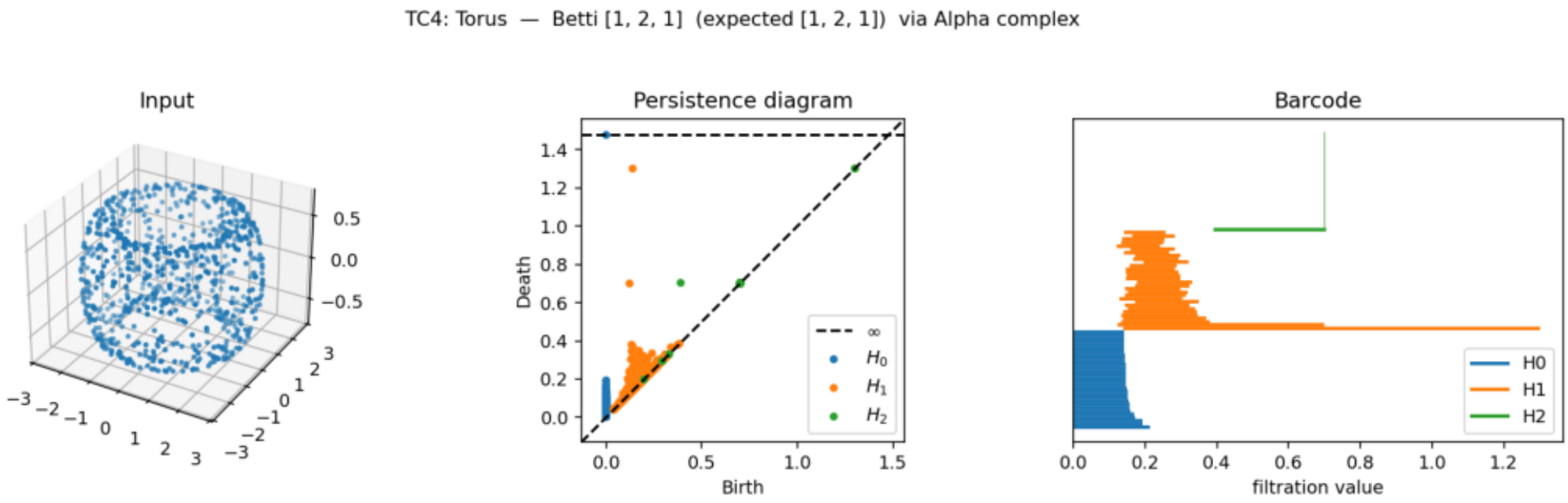
TC2 — input · persistence diagram · barcode



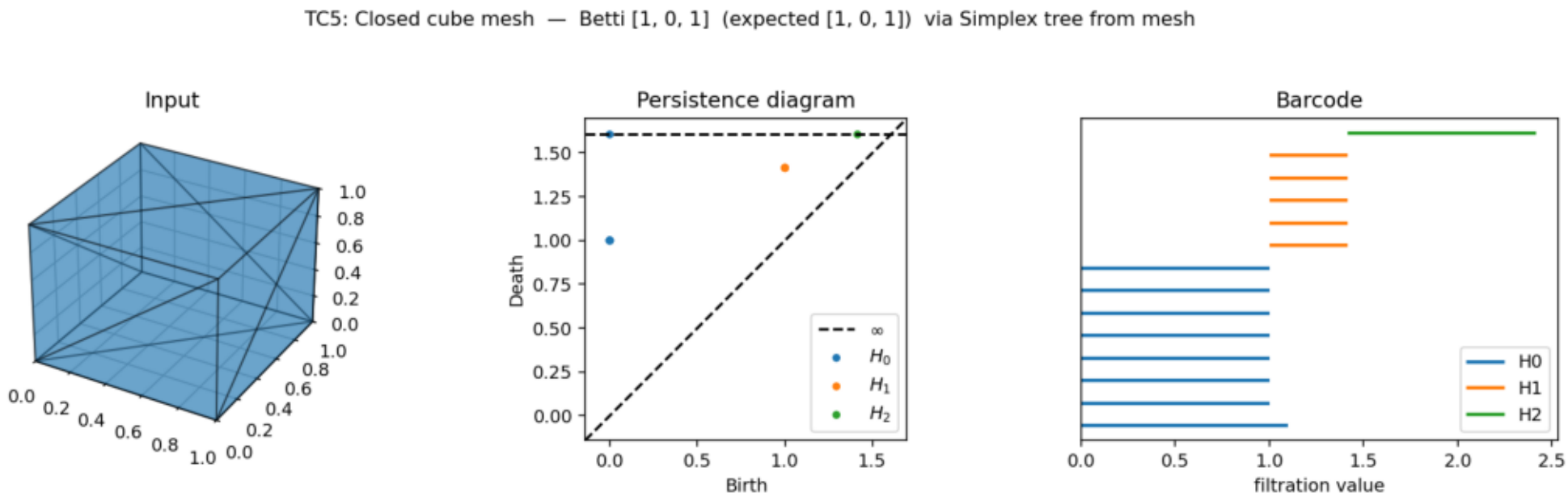
TC3 — input · persistence diagram · barcode



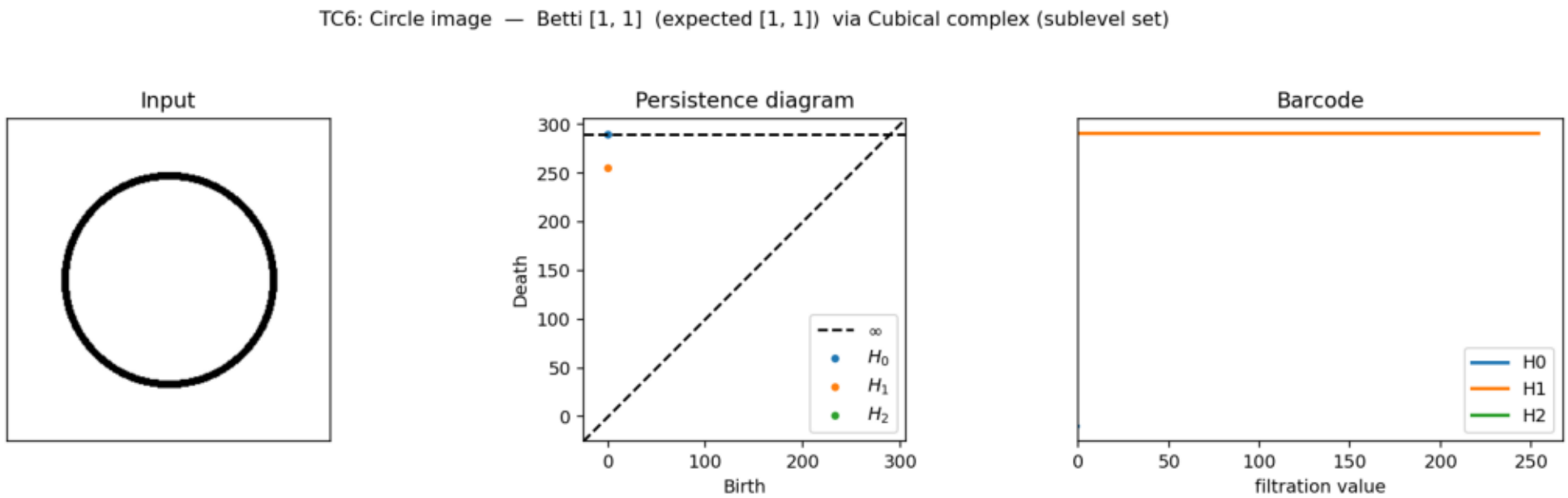
TC4 — input · persistence diagram · barcode



TC5 — input · persistence diagram · barcode



TC6 — input · persistence diagram · barcode



3. Analysis

TC1 (noisy circle): 100 points around the unit circle with 10% Gaussian noise. H1 shows one long bar (the loop) clearly separated from the diagonal cluster of merge bars in H0.

TC2 (annulus): 200 points sampled from a square with a circular hole. Alpha complex (which equals Cech in 2D) gives an H1 bar of persistence 0.59, four times longer than the next-largest noise bar — a clean separation.

TC3 (sphere): the H2 cavity appears at filtration ~0.46 and dies at ~1.64; persistence 1.18 dwarfs all H1 features (largest 0.26). Note: even though Rips eventually fills the interior with essential 3-cells, with maxdim=2 we never insert them, so the cavity stays alive long enough to be unambiguous.

TC4 (torus): Alpha complex produces two long H1 bars (persistences 1.16 and 0.58), corresponding to the two independent torus loops, plus one H2 bar of persistence 0.31 for the interior cavity. The Alpha filtration eventually fills the convex hull with tetrahedra, so the torus features are finite — but their relative persistence is still a clear topological signature.

TC5 (cube mesh): the cube surface is a topological sphere (genus 0). The simplex tree filtration gives 7 short H0 merge bars (the 8 vertices joining), 5 short transient H1 bars from the 1-skeleton before the triangles are inserted, and one essential H2 class — the cavity. After the persistence threshold these reduce to beta = (1, 0, 1).

TC6 (image): a thin black ring on white background. Sublevel-set filtration on intensity finds one long H1 bar — the white interior trapped inside the ring — exactly as expected.